3 Fig 3.1 shows an open cube floating in a liquid.

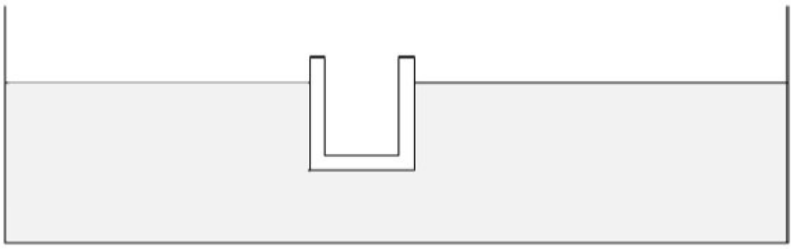


Fig. 3.1

(a) When the cube is displaced downwards by a small distance from the equilibrium position and released. The variation with time *t* of the vertical displacement *y* of the plate is shown in Fig. 3.2.

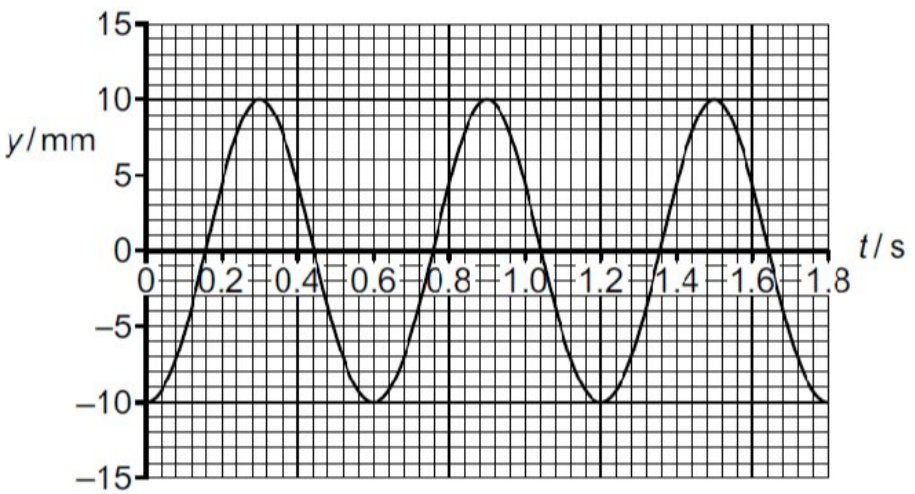


Fig. 3.2

The cube undergoes simple harmonic motion.

The mass *m* of the oscillating cube is 340 g.

(i) Determine the frequency of oscillation.

frequency =..... Hz [1]

(ii) Calculate the total energy of the oscillations.

energy =..... J [3]

(b) In an experiment, surface water waves are generated using a dipper shown in Fig. 3.3. The generated waves are incident on the cube, and produce vertical vibrations of constant amplitude but with a frequency that can be varied.

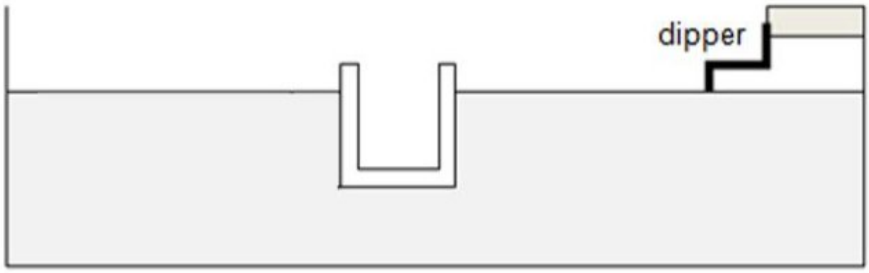


Fig. 3.3

The variation with frequency f of the amplitude y_0 of the oscillations of the cube is shown in Fig. 3.4.

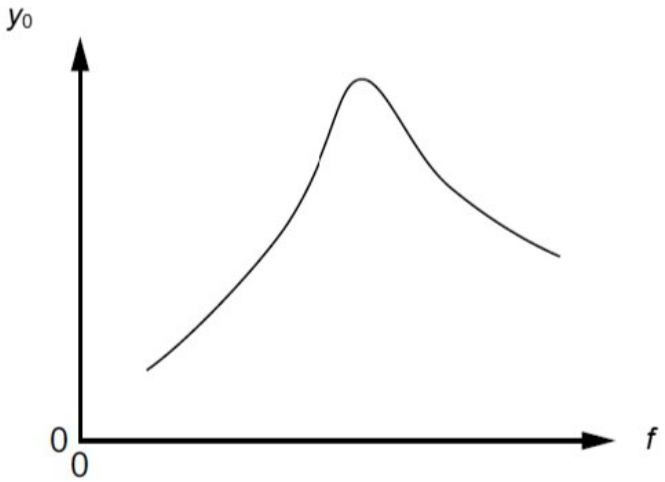


Fig. 3.4

(i) State the name of the phenomenon illustrated in Fig. 3.4.

.....

[1]

(ii) State the frequency at which y_0 is at its maximum.

frequency =..... Hz [1]

(iii) Sketch on Fig. 3.4 the effect on the oscillations if the liquid is more viscous. [2]

[Total: 8]